



Workshop on

Fundamentals of Impedance Control and Haptic Teleoperation for Medical Robotics

Overview

Medical robotics combines advanced control, sensing, and human–robot interaction technologies to enable safer, more precise, and less invasive procedures. This workshop introduces three fundamental technologies that underpin many modern medical robotic systems—**impedance control**, **teleoperation**, and **haptic feedback**—and demonstrates how they work together in real-world applications.

Participants will gain a practical understanding of *impedance control*, which enables robots to regulate their interaction with the environment by controlling the relationship between motion and force. In medical robotics, impedance control can facilitate the realization of features such as virtual fixtures, no-fly zones, and Remote Center of Motion (RCM) constraints by enabling compliant interaction, improving safety, and assisting surgeons during robotic procedures. It also plays a key role in applications such as robot-assisted ultrasound, where maintaining a desired contact force between the ultrasound probe and the patient's body is crucial for acquiring high-quality images.

The workshop will also explore *teleoperation*, the foundational technology behind most surgical robotic platforms, enabling clinicians to remotely manipulate robotic instruments with high precision and dexterity. Beyond clinical use, teleoperation forms the basis of robotic simulators that allow surgeons to practice and refine their skills in realistic virtual environments before entering the operating room.

Finally, participants will be introduced to *haptic feedback*, an emerging capability in medical robotics that allows operators to perceive interaction forces during robotic manipulation. By rendering contact forces to the user, haptic feedback enhances tasks such as tissue palpation, ultrasound scanning, and delicate tissue manipulation, while also providing a more immersive and effective training experience in simulation.

Through demonstrations and hands-on implementation, participants will build a basic teleoperation system incorporating impedance control and haptic feedback, gaining practical insight into the technologies that drive modern medical robotic systems. The workshop is designed for students, researchers, and robotics enthusiasts who wish to bridge the gap between theoretical concepts and real-world applications in surgical and medical robotics.

Software Requirements

- MATLAB & SIMULINK (2024b)
- Blender 3.5

Duration

1.5 Hours

About the Speakers

Dr. Sreekanth Kana received his Ph.D. in Mechanical and Aerospace Engineering from Nanyang Technological University (NTU), Singapore, in collaboration with Rolls-Royce, where his research focused on human-robot collaboration. He subsequently completed postdoctoral research at the Delta-NTU Corporate Laboratory and the Schaeffler-NTU Corporate Laboratory.

He later joined VentureOne, the surgical robotics wing of Karl Storz, where he served as the Technical Lead for Robot Motion Planning and Control. Dr. Kana is currently the Co-founder of ROBOSS, a Singapore-based company developing a medical-grade operating system for surgical robotics with the vision of democratizing access to surgical robotic technology. With over 11 years of experience in robotics research and development, he is the inventor of 4+ patents and the author of 15+ research publications.

Joining him is Shashank Sharma, founder & CEO of ROBOSS, who brings over 20 years of experience in robotics software engineering. At KUKA, he led the development of operating system software supporting more than 160 robot models. He later headed the Software Division at Karl Storz VentureOne, where he led the development of software for next-generation surgical robotic systems. He is the inventor of 10+ patents and the author of 10+ research publications.
